



Complications of Treatment

Interplay between coagulation and inflammation in cancer: Limitations and therapeutic opportunities



Alexander T. Bauer^{a,b}, Christian Gorzelanny^a, Christoffer Gebhardt^a, Klaus Pantel^c, Stefan W. Schneider^{a,*}

^a Department of Dermatology and Venereology, University Medical Center Hamburg-Eppendorf, Hamburg, Germany

^b Department of Dermatology, Venereology and Allergology, Medical Faculty Mannheim, University of Heidelberg, Mannheim, Germany

^c Department of Tumor Biology, University Medical Center Hamburg-Eppendorf, Hamburg, Germany

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ABSTRACT

Advances in understanding the molecular mechanisms of tumor progression have achieved impressive progress in the treatment of cancer and so-called immune checkpoint inhibitors (ICIs) have revolutionized cancer therapy. Indeed, antibody-based drugs blocking immune escape of tumor cells by modulation of T cell responses are increasingly utilized for a wide range of tumor entities. Nonetheless, response rates remain limited, and the development of secondary resistance is a common problem. In addition, by increasing the immune response a variety of severe side effects are provoked. Next to autoimmune responses, activation of the complement system and skin toxicity, an increased incidence for thrombotic complications has been observed associated with an increased mortality rate. Based on this, it can be postulated that the interplay of coagulation with inflammation in the tumor microenvironment is relevant for each step in the tumor life cycle. This review focuses on the coagulation as central player fostering mechanisms associated with tumor progression. Thus, a better understanding of the molecular pathways involved in the complex interaction of circulating tumor cells, the plasmatic coagulation and immune cells may help to improve therapeutic concepts reducing mortality and morbidity associated with cancer.

Introduction

A deepening of understanding the molecular pathways of tumor progression have been the basic requirements for the development of so-called targeted therapies and immune checkpoint inhibitors (ICIs). This new generation of drugs is designed to specifically block tumor cell proliferation, angiogenesis or immune escape mechanisms and intend to reduce high cytotoxicity associated with chemotherapy. On the one hand, much attention has been devoted to the notion that a growing tumor requires blood supply and allowed the development of monoclonal antibodies and small molecule inhibitors directed against angiopoietin-2 (Ang-2), vascular endothelial growth factor-A (VEGF-A), growth factor receptors or their downstream signaling [1]. On the other hand, the therapeutic support of anti-tumor immune responses with immunotherapies led to profound changes in the treatment of cancer

patients [2,3]. The most noticeable results have been achieved with ICIs involving monoclonal antibodies against the cytotoxic T-lymphocyte associated protein 4 (CTLA-4), the programmed death receptor 1 (PD-1) or its ligand (PD-L1) [4]. Clinical trials using nivolumab or pembrolizumab, humanized antibodies to PD-1, provide response rates of more than 40% in patients with lung cancer, melanoma and renal carcinoma among others demonstrating the highest anti-tumor activity of therapeutic approaches during the last decade [5]. However, despite considerable progress in the treatment of cancer using targeted therapies and immune checkpoint inhibition, limited response rates, acquired resistance and toxic side effects remain challenging [2,6]. In this context it has been observed that patients treated with ICIs have a higher risk for thrombotic complications [7,8]. Indeed, so-called cancer-associated thrombosis (CAT) is a recognized complication in malignancy and it was demonstrated that the risk for the development of CAT is 9-fold higher in

Abbreviations: VWF, von Willebrand factor; ICI, immune checkpoint inhibitor; ADAMTS13, a disintegrin and metalloprotease with thrombospondin type 1 motifs, member 13; CAT, cancer-associated thrombosis.

* Corresponding author at: Department of Dermatology and Venereology, University Hospital Hamburg-Eppendorf, Martinistrasse 52, 20246 Hamburg, 68167, Germany.

E-mail address: st.schneider@uke.de (S.W. Schneider).

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cancer patients than in the general population [9,10]. Those studies indicate that the prevalence of CAT depends mainly on the tumor entity, the disease stage, the tumor grade, location of metastases, the body mass index and the anti-tumor therapy [11,12]. The risk to develop venous thromboembolism is higher in cancer patients receiving chemotherapy or targeted therapy [9]. Further, cancer patients under immune checkpoint therapy are at high risk for the development of CAT associated with increased mortality rates [7,13,14]. However, the molecular mechanisms of how a dysregulation of the inflammation mediated by immune checkpoint inhibition affects the hemostatic system has not been properly addressed so far.

This review summarizes tumor-specific pathways of thrombosis and inflammation with the vascular endothelium as key player located at the interface between coagulation, inflammation, and metastases. A precise understanding of how endothelial cells drive inflammation and thrombosis is essential for the design of novel strategies increasing the efficacy of immune checkpoint inhibition and targeted therapies (Fig. 1).

Tumor-associated coagulation

The regulation of hemostasis is a complex and dynamic process controlled by the plasmatic coagulation, circulating platelets and the vascular endothelium essential for the balance between clot formation and bleeding. Evidence suggests that cancer cells can cause an imbalance of this equilibrium by activation of blood clotting [15]. Importantly, thrombotic complications are not only correlated with disease progression and prognosis [9], but may also indicate an occult cancer [16]. These findings can be explained by tumor cell-associated procoagulants, which allow the binding and activation of coagulation factors on the surface of tumor cells and end up in coagulation abnormalities [15]. Of those, tissue factor (TF) expressed on the malignant cell surface,

can play a central role in cancer-associated coagulation [17]. A high TF expression in tumor tissue of cancer patients has been correlated with tumor progression, worse prognosis and thrombosis [18]. In response to blood exposure, TF interacts with plasmatic coagulation factors and promotes the generation of thrombin, which has multiple cellular effects. Thrombin exerts its thrombotic effects via cleavage of fibrinogen and exhibits pleiotropic cellular effects mediated through protease activated receptors (PARs). Thrombin cleaves and activates PAR-1, -3 and -4 expressed on the surface of many cells, including neurons, fibroblasts, immune cells, platelets and endothelial cells [17]. Cancer cells overexpressing PARs result in enhanced metastasis in mouse models [19]. Thus, TF on the cell surface could represent a mechanistic interconnection between hemostasis and tumor progression. The activated receptors are coupled to G-proteins and induce several cellular reactions, such as cell shape changes, migration and the secretion of growth factors. However, focusing on its biological functions it is difficult to draw mechanistic conclusions how the thrombin-PAR-axis promotes coagulation and tumor progression. Studies have demonstrated that thrombin activates human platelets via PAR-1 and PAR-4, resulting in the secretion of a wide range of molecules [20]. Because platelets secrete factors including matrix metalloproteinases, platelet factor 4 and VEGF, tumors may use the proangiogenic properties of platelets for the formation of new blood vessels, supporting the migration and the activation of endothelial cells [17,20]. On thrombin-mediated PAR activation, endothelial cells undergo several cellular changes including migration and the upregulation of VEGF-A, which are required for angiogenesis [20,21]. Both, clot formation and the extravasation of tumor cells, take place at the endothelial surface. Endothelial cells play a pivotal role in controlling adhesion and transmigration of immune and cancer cells circulating in the vascular system. It is plausible that the communication of a tumor cell with the vascular endothelium represents

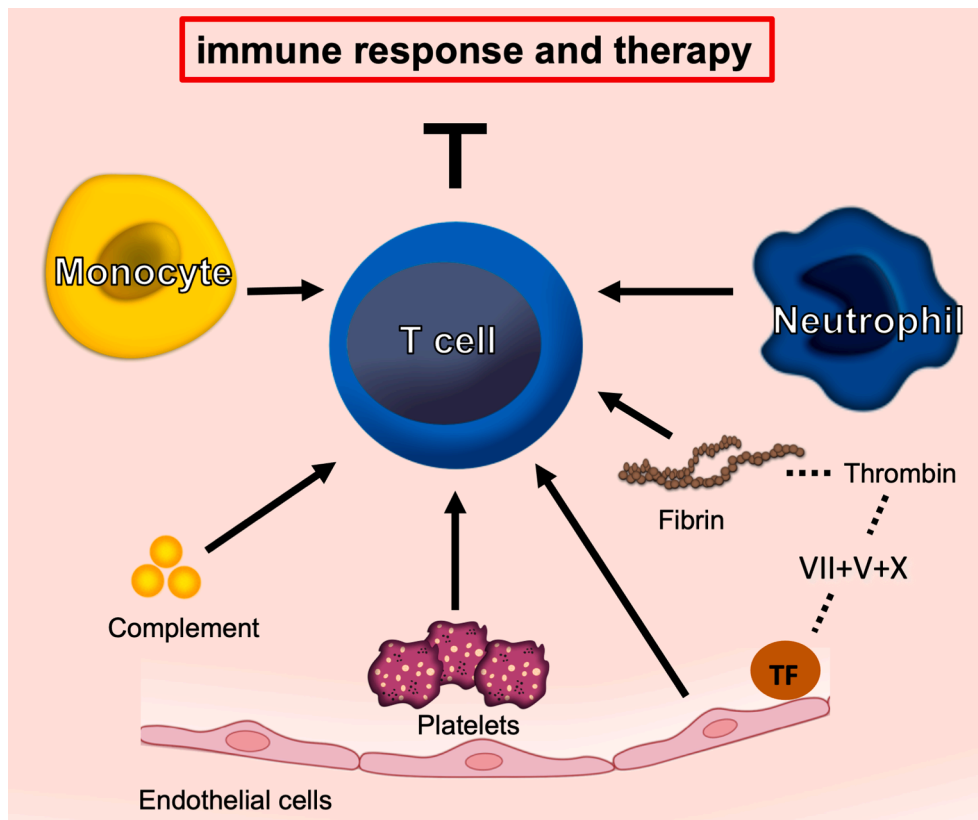


Fig. 1. Interplay between cancer-related immunothrombosis and responses to immune checkpoint inhibition. The formation of a cancer-related “immunothrombosis” is orchestrated by complex cellular interactions. Of those, endothelial cells, platelets, monocytes and neutrophils, the plasmatic coagulation activated by tissue factor (TF) and complement factors can alter T cell responses leading to resistance and severe side effects associated with immune checkpoint inhibition.

initial steps in extravasation and subsequent metastasis formation. In principle, tumor cell-mediated endothelial cell activation transforms the non-thrombotic and anti-inflammatory cell surface into an adhesive and procoagulant surface [22]. This process is mediated via tumor cell-generated thrombin or tumor cell-secreted VEGF-A, which promotes endothelial cell activation characterized by the secretion of several factors stored in endothelial Weibel-Palade bodies, such as interleukin-8, Ang-2, P-selectin and von Willebrand factor (VWF) [22,23]. Upon secretion, the hemodynamic protein VWF is stretched within the blood flow exposing binding sites for circulating platelets, thus promoting its procoagulant activity [24]. Therefore, it is not surprising that molecules stored in endothelial cells next to VWF have been investigated for the prediction of CAT. Effectively, elevated plasma levels of soluble P-selectin [25] and VWF [26] as biomarkers for cancer patients could be of great use for clinicians. Using different mouse models, it was previously shown that VWF secreted from activated vessels within a primary tumor promote platelet aggregation associated with thrombotic vessel occlusions [23,27]. However, tumor vessels are disorganized and this raises the question whether conclusions drawn in primary tumor tissue can directly be transferred to distant organs with metastases but regular vessels. To answer this question, a mouse model based on intravenous injection of tumor cells was applied. Importantly, the number of metastatic foci in the lungs is directly correlated to the formation of VWF – platelet aggregates indicative for similar mechanisms of endothelial cell activation and VWF secretion [27]. Moreover, intradermal inoculation of tumor cells followed by the induction of lymph node metastases resulted in a pronounced VWF network formation and thrombotic vessel occlusions in metastasis-free brains, livers and lungs [27]. Based on this observation the concept arises that a primary tumor mediates the stimulation of the vascular endothelium by tumor-derived factors, such as TF and VEGF-A, resulting in the secretion of several factors stored in endothelial cells and platelets. As it is obvious that coagulation triggers tumor progression, factors of the coagulation cascade have been identified as promising biomarkers in malignancy. Considering the critical role in the development of CAT, it is not surprising that increased levels of d-dimers and VWF are associated with cancer onset and progression [28,29]. However, while the importance of the hemodynamic activation of VWF for platelet binding and aggregation may provide a plausible explanation for tumor cell dissemination and the formation of intravascular thrombi as prerequisite for the development of thrombosis, it remains to be established how thrombotic events communicate with inflammatory processes.

Tumor-associated inflammation

Several studies report increased levels of circulating TF in patients with cancer [18]. It is generally accepted that TF-mediated thrombin generation mediates platelet activation that is crucial for hypercoagulation in malignancy [18]. Importantly, thrombin can also activate endothelial cells triggering the exposure of E-selectin, P-selectin, and VWF [22]. Both, E-selectin and P-selectin facilitate the adhesion of circulating tumor cells promoting metastasis [30,31]. From the inflammatory point of view, selectins are the initial mediators for the recruitment of leukocytes to the vessel wall [32]. In addition to selectins, VWF multimers are exposed, which form adhesive networks under pathophysiological conditions [24,33]. Indeed, experimental studies reveal that VWF networks act as linker for adhesion of monocytes and neutrophils to the vascular endothelium supporting extravasation of leukocytes and therefore inflammation [34,35]. Of those, neutrophils seem to play a critical role for malignancy, coagulation and inflammation. Depletion of neutrophils inhibits coagulation [36] and affects metastasis [37]. Neutrophils recruited to organs distant to the primary tumor appeared to shape the pre-metastatic niche [37]. The granulocytes secrete several growth factors such as VEGF-A, platelet-derived growth factor (PDGF) and tumor growth factor- β (TGF- β) which promote angiogenesis and initiate metastasis formation by facilitating the

adhesion of circulating tumor cells [38]. Neutrophil extracellular traps (NETs) composed of DNA/histone strings and immune modulatory factors may promote the proliferation of tumor cells and alter the inflammatory milieu by modulation of T cell responses [39,40]. Interestingly, it was also shown that NETs are able to directly trap circulating tumor cells and support extravasation and thus metastasis [41]. NETs are known to potentiate thrombosis through several mechanisms and they are major components of thrombi [42,43]. It was shown that NETs bind to VWF networks and platelets [44], promote fibrin generation via the coagulation factor XII [36] and enhance TF activity [44]. Clinical trials provide evidence that elevated levels of NETs can be used as biomarkers to detect early tumors and predict the development of metastases [45,46]. Moreover, NETs are well-known risk factors for tumor-associated vascular dysfunction, thrombosis and organ-failure, and several clinical studies suggest that NETs are likely to contribute to CAT [47,48]. Thus, the close interaction between the innate immune system and the coagulation suggests overlapping mechanisms of hemostasis and inflammation in cancer and was previously described by the term “immunothrombosis” [49].

Tumor-associated complement function

Recently, the complement system as part of the innate immune response has gained increased attention as major contributor to cancer progression [50]. The complement system is composed of a cascade of self-activating proteases and its mechanistic organization is reminiscent of the composition of the plasmatic coagulation system and its multi-component enzyme complexes. Three major activation routes can be defined: (a) the classical pathway, (b) the mannose-binding lectin pathway and (c) the alternative pathway [51]. All activation pathways culminate (i) in the generation of the cofactor and opsonin C3b, which is covalently bound to complement-attacked surfaces, (ii) the formation of the anaphylatoxins C3a and C5a and finally (iii) the formation of the lytic membrane attack complex (MAC; C5b-9) [51]. Traditionally, the complement system is linked to the elimination of pathogens, immune cell recruitment and initiation of the adaptive immune response [52,53]. However, growing body of evidence suggests the complement system to be involved in the generation of a tumor-promoting environment, for instance by leukocyte attraction and manipulation of their phenotype [54].

Deficiency of C3 and C5 in mice resulted in reduced venous thrombosis, further suggesting a role of complement activation in coagulation and thrombosis [55]. However, mechanisms by which the complement cooperates with the coagulation are poorly understood. For example, the hemodynamic protein VWF may provide a platform for binding and activation of complement products and promotes microvascular vessel occlusions [56]. The complement factor C3 participates in thrombosis via activation of platelets [55] and it was previously demonstrated that thrombin is involved in the cleavage of C5 to C5a [57]. Accordingly, the generated anaphylatoxin C5a regulates TF expression on the vascular endothelium *in vitro* and on isolated neutrophils, a process that may further link immunity to coagulation [58,59]. From this scenario the concept appears that complement activation may promote “immunothrombosis” under pathologic conditions as manifested in malignancy. Distinct studies reported complement activation in human cancers, such as ovarian cancer and brain tumors [60,61] and complement activation was proposed to serve as a prognostic marker for patients with lung tumors and colorectal cancer [62,63]. Notably, upregulation of the complement C3a receptor was shown to activate neutrophils followed by induction of NETosis and thrombosis and to promote tumor progression [64]. In a preclinical mouse model for lung cancer, the simultaneous inhibition of PD-1 and C5a strongly reduced metastases and improved the efficacy of anti-PD-1 treatment [65]. This clearly reveals a potential therapeutic benefit of anti-complement drugs for cancer-associated inflammation, thrombosis and metastasis. Although a clinical study showed that inhibition of C5 with the humanized antibody

eculizumab reduced the risk for VTE in patients with paroxysmal nocturnal hemoglobinuria, it has to be established if similar mechanisms are involved in cancer [66]. However, to the best of our knowledge, no ongoing clinical trials evaluate the therapeutic benefit of the C5 inhibitor eculizumab in tumor patients.

Coagulation and inflammation in response to checkpoint inhibition

Fundamental discoveries in understanding immune escape mechanisms of tumors have revolutionized cancer therapy during the last decade. To escape from immune surveillance, tumors follow multiple strategies to block T cell-mediated immune responses [3]. Briefly, one of the most important molecules, known to inhibit both T cell proliferation and interleukin-2 production during early stages of immune responses, is the cytotoxic T-lymphocyte associated protein 4 (CTLA-4). In later stages of immunity, the protein programmed death-1 (PD-1) expressed on immune cells and the programmed death ligand-1 (PD-L1), which is expressed on the surface of tumor cells, block T cell functions [2]. ICIs are antibody-based drugs, which block ligand / receptor binding of molecules such as CTLA-4 and PD-1 important for the suppression of T cell activation and function. However, despite the enormous clinical success of ipilimumab directed against CTLA-4, nivolumab, pembrolizumab or cemiplimab targeting PD-1 and avelumab blocking PD-L1 [67], limited response rates and severe side effects are still challenging [3,5,67].

A low T cell infiltration characterizing the so-called “cold” tumors is a big challenge for immunomodulatory therapy. A defective priming and activation of T cells may also influence the efficacy of checkpoint inhibition [68]. Because it is difficult to provide mechanistic insights, which may explain these therapeutic limitations, basic research is required to understand the interplay among hemostasis, immune responses and tumor progression. Clinical data provide evidence for an association between ICIs and thrombotic complications. Several studies describe the occurrence of the Trousseau's syndrome, arterial thrombosis, multiple organ thromboembolism and disorders of the coagulation-fibrinolysis system in patients receiving ICIs [69–72]. A large pharmacovigilance study analyzed almost 30 million records including more than 100,000 ICI related cases over a timeframe of five years originating from the US Food and Drug Administration Adverse Event Reporting System (FAERS) database [73]. Here, a strong correlation between ICIs and an increased incidence of coagulatory adverse events like disseminated intravascular coagulation, hypercoagulation and *purpura fulminans* was detected. Moreover, a retrospective cohort study of 228 patients with malignant melanoma receiving ICI, described a 12-month cumulative incidence rate for venous thromboembolism of 21.3% [13]. In strong accordance, a large study including 672 patients with different tumor entities observed a cumulative incidence for venous thromboembolism of 12.9% for malignant melanoma patients [7]. Because the occurrence of thrombotic events was associated with an increased mortality in the same study, one may speculate that a procoagulatory milieu supports immune suppressor properties mediated by a tumor. However, the molecular mechanisms of the interaction of immune checkpoint inhibition and CAT are not known. This highlights the need for a deep mechanistic evaluation of the crosstalk between coagulation and inflammation in cancer. It was recently shown that the TF–thrombin–PAR-1 axis in pancreatic tumor cells mediates suppression of immunity by modulation of CD8⁺ cytotoxic T cells [74]. Another report demonstrates that Factor X mainly produced by macrophages and monocytes promotes immune evasion of tumor cells [75]. Anti-coagulation with rivaroxaban not only enhanced the recruitment of cytotoxic T cells to the tumor site, but also improved anti-PD-1 therapy in a mouse tumor model [75].

Next, circulating platelets are not only deformable bags recruited by VWF to plug damaged vessels and to avoid bleedings [76]. In fact, platelets express PD-1L, which allows a direct interaction with T cells

and might affect the efficacy of immune checkpoint therapies [77]. It has been shown that T cells bound to platelets exhibit a reduced proliferation and activity [78]. This could correlate with a significant reduction in the efficacy of immune checkpoint therapy using antibodies against PD-1 and suggests a direct role of platelets in cancer-related immune escape. Next to pro-tumorigenic effects mentioned above, there is clear evidence that thrombin cleaves glycoprotein A repetitions predominant (GARP) on platelets, leading to the liberation of platelet-derived TGF- β a potent suppressor of T cell proliferation, activation and effector functions [79,80]. Accordingly, inhibition of platelets using aspirin and clopidogrel enhanced the therapeutic benefit of immunotherapy in a mouse model for malignant melanoma [81]. Among a plethora of platelet-derived molecules, TGF- β has been shown to reduce cytotoxicity of natural killer cells [82], to promote tumor evasion by blocking cytotoxic T cells and to induce therapy resistance [83]. Recently, M7824, a fusion protein targeting PD-L1 and TGF- β was tested in a phase I clinical trial suggesting therapeutic benefit in patients with advanced solid tumors [84,85].

Activated platelets also trigger innate immune responses via neutrophil recruitment and NETosis [86]. The inhibition of anti-tumor CD8⁺ T cell responses is modulated by neutrophils through several mechanisms. Neutrophils are able to block cytotoxic T cell functions via the expression of iNOS and the production of NO [87]. It was shown that neutrophils express PD-1L associated with interferon-dependent induction of T cell apoptosis via binding to PD-1 [88]. When neutrophils were depleted in a mouse model for pancreatic adenocarcinoma, T cell infiltration into the tumor and metastases was increased and an improved response to PD-1 inhibition was observed [89]. In addition, a high neutrophil-to-lymphocyte ratio in the blood predicts a poor overall survival of cancer patients [90]. Especially in malignant melanoma patients, high neutrophil counts correlate with poor prognosis and a limited response to checkpoint inhibition [91]. Further studies using murine tumor models demonstrated that the receptor tyrosine kinase c-MET link T cell responses and neutrophil action during tumor progression. Treatment with the kinase inhibitor capmatinib blocked the recruitment of neutrophils into tumors, impaired immunosuppressive properties of neutrophils, promoted T cell infiltration, and enhanced treatment efficacy of immunotherapies [92]. In support of this concept, clinical studies have demonstrated beneficial therapeutic effects of small molecule c-Met inhibitors as monotherapy or in combination with other drugs [93]. Owing to the incomplete understanding of the adaptive T cell responses in malignancy, mechanisms of resistance to ICIs are not fully understood. In the process of tumor progression, T cell responses are modulated by numerous cellular interactions within the tumor microenvironment (Fig. 2).

A further functional bridge between neutrophils and the adaptive arms of immune responses is represented by the contribution of the complement. Focusing on the concept of “immunothrombosis”, neutrophils, circulating thrombocytes and activated endothelial cells, provide platforms for complement binding and activation [94] and are able to modulate adaptive T cell responses [95]. Mice deficient in C3 and C5 have reduced experimental venous thrombosis [55]. High levels of C3 are associated with an elevated risk for the development of VTE in humans [96]. In this regard it should be mentioned that C5a is a potent mediator of immunosuppression in the tumor microenvironment. Genetic deficiency or pharmacological inhibition of the C5 Receptor (C5R) increased CD8⁺ T cell numbers [97]. Thus, the crosstalk of complement and coagulation cascades may not only play a crucial role in thrombosis and inflammation, but also may regulate thrombo-inflammatory processes in tumor progression.

In line, the simultaneous blockage of PD-1 and C5a in an animal model for lung metastasis formation increased the anti-metastatic potential of PD-1 inhibition [65]. These findings were the rational basis for a clinical phase I study combining treatments with the C5a receptor antagonist (IPH5401) and an antibody against PD-1 ligand (durvalumab) in tumor patients (NCT03665129) and highlight the clinical

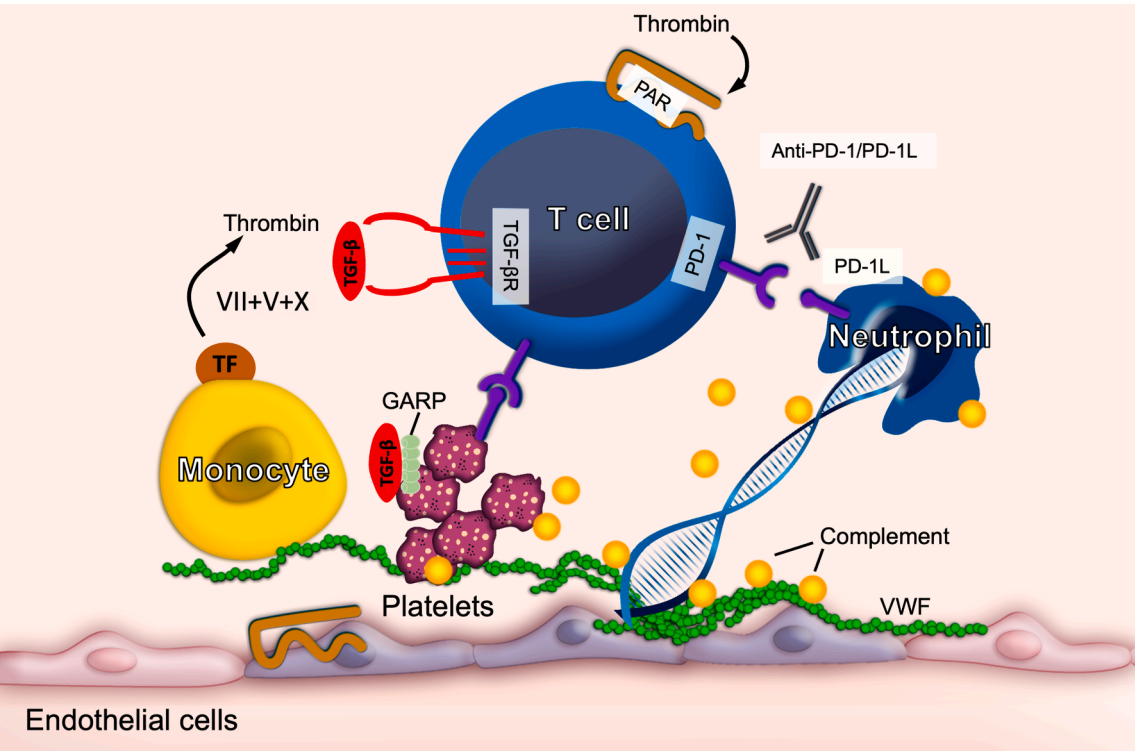


Fig. 2. Central role of the vascular endothelium in the interaction between coagulation and inflammation. A tumor has different prothrombotic pathways for immune evasion and to counteract targeted inhibition of programmed death receptor 1 (PD-1) or its ligand (PD-1L). In response to cancer-related exposure of tissue factor (TF), the vascular endothelium gets activated. The secretion of P-selectin and the formation of luminal von Willebrand factor (VWF) networks mediate the recruitment and activation of circulating monocytes, neutrophils and platelets. TF exposed by monocytes promotes the generation of thrombin, which cleaves and induces protease-activated receptor (PAR)-mediated signal transduction in T cells. Platelets mediate immune suppression through different pathways. Thrombin cleaves the glycoprotein A repetitions predominant (GARP) receptor on the platelet surface required for the activation and liberation of transforming growth factor- β (TGF- β). In addition, PD-1L is present on platelets, which presumably affects T cell activity by binding to PD-1 on the surface of T cells. Next, exposure of PD-1L by neutrophils was demonstrated to induce T cell apoptosis. In addition, neutrophils secrete neutrophil extracellular traps (NETs) consisting of DNA and histones with strong binding affinity to VWF strings, TF and complement factors. The activated complement factors C3a and C5a trigger thrombosis by activation of platelets and modulate inflammation by impairment of CD8⁺ T cell functions.

potential of targeting the complement in the context of immune checkpoint inhibition [98].

Future perspective and possible treatment strategies

Tumor patients show a procoagulatory state followed by an increased incidence of CAT associated with pronounced tumor progression, a lower treatment benefit and a worse overall survival. Up to now, *in vitro* and *in vivo* data suggest that a procoagulatory milieu supports tumor immune escape and interferes with immune therapy. One may speculate that a synergistic therapeutic regimen targeting mechanisms of tumor-mediated coagulation and tumor-induced immune escape could be beneficial for tumor patients. Given the multidirectional nature of the tumor microenvironment axis suggests that therapeutic approaches to suppress one of these pathways could be considered to increase treatment response rates and to reduce toxic side effects (Table 1).

Combination therapies with repurposed drugs already in clinical use to overcome limitations and resistance to immune checkpoint inhibition represent a very promising treatment strategy for cancer in the near future (Fig. 3). Of those, the antibody eculizumab is already in clinical use for the treatment of patients with paroxysmal nocturnal hemoglobinuria (PNH) [99] and atypical hemolytic uremic syndrome (aHUS) and reduces the activation of the complement, thrombosis and endothelial damage [100]. Next to the blockage of tumor progression through complement inhibition also the modulation of neutrophil activity appears to be an attractive target in tumor therapy. In line, patients with cystic fibrosis are already treated with DNase I to degrade

Table 1
Inhibitors of the immunothrombosis.

Drug	Target	Clinical indication	References
M7824	fusion protein against PD-L1 and TGF- β	cancer	[84]
Eculizumab	monoclonal antibody against C5	atypical hemolytic uremic syndrome (aHUS); paroxysmal nocturnal hemoglobinuria (PNH)	[99,100]
IPH5401	antibody against C5a receptor	cancer	[98]
Recombinant DNase I	enzymatically cleaves extracellular DNA	cystic fibrosis	[101]
Recombinant ADAMTS13	cleaves and inactivates von Willebrand factor strings	congenital thrombotic thrombocytopenic purpura (TTP)	[105]
Caplacizumab	nanobody against A1 domain of von Willebrand factor activate	acquired thrombotic thrombocytopenic purpura (TTP)	[104]
Low molecular weight heparins (LMWH)	antithrombin III and inhibit thrombin activation	venous thromboembolism, cancer	[10]

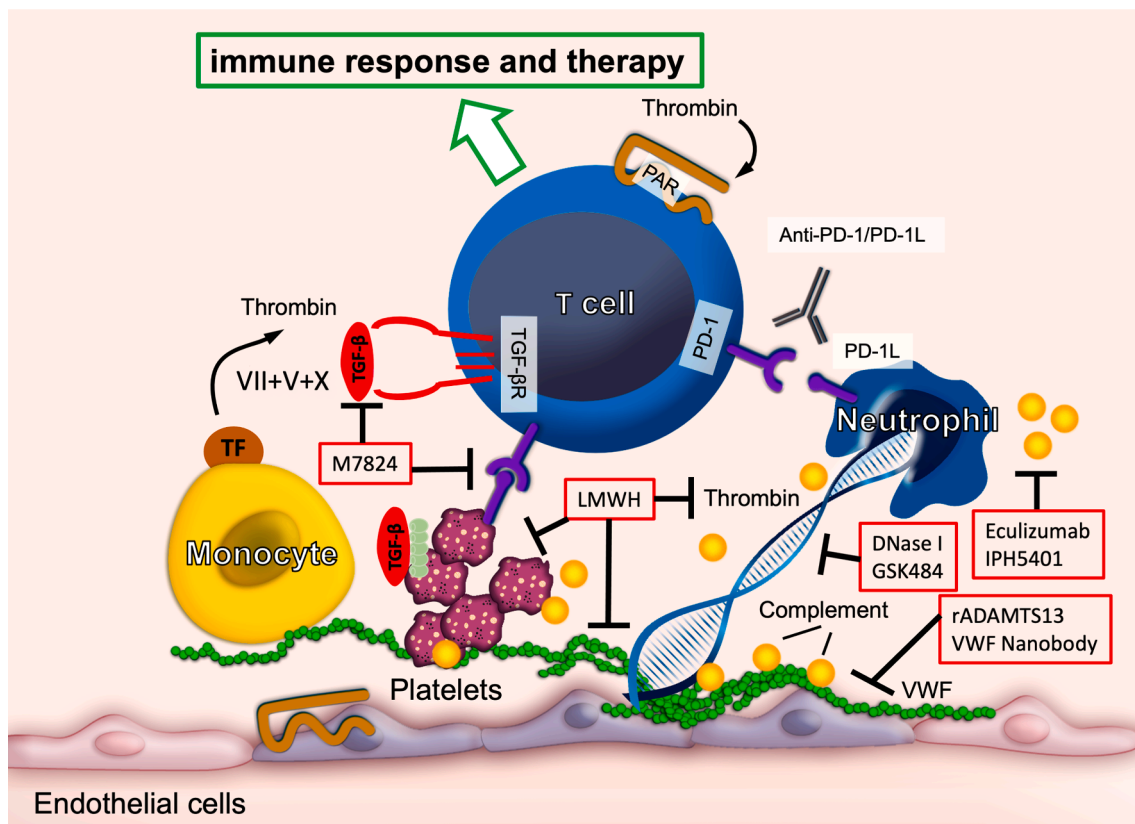


Fig. 3. Therapeutic strategies to support ICIs. The consequence of cancer-related endothelial cell activation is thrombosis. Von Willebrand factor (VWF)-mediated recruitment of monocytes and neutrophils and the aggregation of platelets can be blocked by enzymatic degradation of VWF using recombinant ADAMTS13 (rADAMTS13) and nanobodies targeting VWF (caplacizumab). Caplacizumab binds the A1 domain of VWF and inhibits VWF-mediated platelet binding. Recombinant DNase I and the inhibitor GSK484 degrade extracellular DNA or inhibit the secretion of neutrophil extracellular traps, respectively. Because the fusion protein M7824 is directed against programmed death receptor 1 ligand (PD-L1) and transforming growth factor- β (TGF- β) an inhibition of platelet- and neutrophil-induced blockage of T cell responses can be achieved. Eculizumab and IPH5401 are antibody-based strategies to interfere with complement factor 5 and its receptors. Low molecular weight heparins (LMWH) such as tinzaparin bind and inhibit the vascular endothelial growth factor (VEGF)-A and block thrombin generation. This inhibits platelet aggregation, signaling of protease-activated receptors (PARs) and the secretion of the procoagulant VWF.

extracellular NETs without significant side effects [101]. Notably, NET formation could be reversed by treatment with DNase I or by blocking peptidyl arginine deiminase 4 (PAD4) with the inhibitor GSK484 [102,103]. Also, the inhibition of hypercoagulation through the administration of an anti-VWF nanobody that blocks the A1 domain and thus VWF-mediated platelet binding or treatment with recombinant “a disintegrin and metalloprotease with thrombospondin type 1 motifs, member 13” (ADAMTS13) cleaving and inactivating VWF could represent a treatment option to support immune checkpoint inhibition. The anti-VWF nanobody caplacizumab is used to treat acquired thrombotic thrombocytopenic purpura (TTP) [104]. The von Willebrand factor-cleaving protease ADAMTS13 cleaves and inactivates VWF fibers within seconds [33] and recombinant ADAMTS13 has been used to treat patients with severe congenital ADAMTS13 deficiency showing a good toleration and safety profile [105].

A similar approach to prevent cancer-related thrombosis would be the treatment with low molecular weight heparins (LMWHs). Anticoagulation with heparin is already in clinical practice to prevent venous thrombosis in cancer [10]. Apart from its anticoagulant activity, heparins possess a variety of activities which could interfere with tumor progression. Of those, blocking P-selectin, VEGF-A, inhibition of heparanase, destabilization of NETs and inhibition of VWF secretion could represent important issues to be considered when designing therapies targeting inflammation, impaired vessel perfusion and cancer [23,106,107].

Conclusions

The development of immunotherapy in multiple cancers including melanoma has revolutionized oncology. However, mainstream initiation of immune checkpoint therapy to treat cancers is obstructed by difficulties in the prediction of immune-related adverse side effects and limited response rates in some cancer patients. This apparent discrepancy between the therapeutic success and therapy resistance emphasizes the need of basic translational research for a better understanding of the crosstalk between tumor cells, thrombosis and inflammation, and the dynamic nature of these cellular networks in the context of immune evasion and metastases. Indeed, developing strategies to selectively target the coagulation and simultaneously affect complement factors and inflammatory responses to promote anti-tumor effects require further research to fully understand the interaction of these pathways in the tumor microenvironment. With this in mind, the targeting of coagulation as bridge between the innate and adaptive immune system could provide a novel strategy to overcome treatment resistance in response to checkpoint inhibition. In this context, monitoring tumor evolution by sequential analysis of circulating tumor cells or cell-free DNA can help to monitor the efficacy of combination therapies [108,109].

CRediT authorship contribution statement

A.T.B.: Conceptualization, Validation, Visualization, Writing-original draft. C.G., C.G. and K.P.: Writing-review & editing. S.W.S.: Conceptualization, Supervision, Writing-review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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